

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	2 MARKS

3.2 – Solution Requirement

The proposed solution aims to provide a comprehensive understanding of electricity consumption patterns while helping consumers and energy providers make informed decisions regarding energy usage and conservation. To achieve this objective, the solution must include several functional and non-functional requirements that support efficient data collection, analysis, monitoring, and reporting.

Functional Requirements

- Collect electricity consumption data from smart meters, utility bills, or energy monitoring systems.
- Store and manage historical electricity usage data securely.
- Analyse consumption patterns based on daily, weekly, monthly, and seasonal trends.
- Identify peak and off-peak electricity usage periods.
- Generate reports and visualizations such as graphs, charts, and dashboards.
- Provide recommendations for reducing electricity consumption and energy costs.
- Monitor the performance of energy-efficient appliances and renewable energy systems.
- Send alerts or notifications when electricity usage exceeds predefined limits.
- Compare consumption patterns across different households or time periods.

Non-Functional Requirements

- Ensure data accuracy and reliability.
- Maintain user privacy and data security.
- Provide a user-friendly and easy-to-understand interface.
- Support real-time monitoring and reporting.
- Ensure system scalability for handling large volumes of consumption data.
- Deliver fast and efficient data processing and analysis.
- Maintain high system availability and reliability.

Technical Requirements

- Smart meters or electricity monitoring devices for data collection.
- Database system for storing consumption records.
- Data analytics tools for pattern identification and forecasting.
- Dashboard or mobile application for visualization and monitoring.
- Internet connectivity for real-time data transmission.
- Integration capabilities with renewable energy systems and smart home devices.

User Requirements

- Easy access to electricity consumption information.
- Clear visualization of energy usage trends.
- Personalized energy-saving recommendations.
- Real-time monitoring of electricity consumption.
- Alerts for unusual or excessive energy usage.
- Information about potential cost savings and environmental benefits.